

KNOWLEDGE

1. Evaluate the following limits.

a. $\lim_{x \rightarrow 2} \frac{x^2 + 6x - 16}{|x-2|(x+3)}$

$$f(x) = \begin{cases} \frac{(x+8)(x-2)}{(x-2)(x+3)} = \frac{x+8}{x+3}, & x > 2 \\ \frac{(x+8)(x-2)}{-(x-2)(x+3)} = \frac{-(x+8)}{x+3}, & x < 2 \end{cases}$$

$$\lim_{x \rightarrow 2^-} f(x) = \frac{-(2+8)}{2+3} = \frac{-10}{5} = -2$$

$$\lim_{x \rightarrow 2^+} f(x) = \frac{2+8}{2+3} = \frac{10}{5} = 2$$

$\therefore \lim_{x \rightarrow 2} f(x)$ **DNE** because $\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$

b. $\lim_{x \rightarrow 9} \frac{3-\sqrt{x}}{27-\sqrt{x^3}}$

$$\lim_{x \rightarrow 9} \frac{3-\sqrt{x}}{27-\sqrt{x^3}} = \lim_{x \rightarrow 9} \frac{3-x^{\frac{1}{2}}}{27-x^{\frac{3}{2}}}$$

$$\lim_{a \rightarrow 3} \frac{3-a}{27-a^3} = \lim_{a \rightarrow 3} \frac{3-a}{(3-a)(9+3a+a^2)} = \lim_{a \rightarrow 3} \frac{1}{a^2+3a+9} = \frac{1}{9+9+9} = \frac{1}{27}$$

$a = x^{\frac{1}{2}}$
 $x = a^2$

(2)
(3)

2. Determine the value(s) of x for which the following function is discontinuous. Identify the type of discontinuity. Justify your response.

$$f(x) = \begin{cases} x^2 + 7x - 5, & \text{if } x \neq 1 \\ 2, & \text{if } x = 1 \end{cases}$$

Type: Point discontinuity
 $x = 1$

If you plug in 1 to the quadratic, you get a y value of 3 ($1^2 + 7(1) - 5 = 1 + 7 - 5 = 3$), however $f(x)$ has a y value of 2 when x is 1, meaning there is a removable discontinuity at $x=1$, because the rest of the function is a polynomial, the rest is continuous.

3. Determine the equation of the tangent of the function, $f(x) = \frac{2}{\sqrt{x+5}}$ at $x = 4$.

$$f(4) = \frac{2}{\sqrt{4+5}} = \frac{2}{\sqrt{9}} = \frac{2}{3} \quad f(4+h) = \frac{2}{\sqrt{4+h+5}}$$

$$f'(4) = \lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h} = \lim_{h \rightarrow 0} \frac{\frac{2}{\sqrt{4+h+5}} - \frac{2}{3}}{h} = \lim_{h \rightarrow 0} \frac{\frac{6 - 2\sqrt{4+h+5}}{3\sqrt{4+h+5}}}{h} = \lim_{h \rightarrow 0} \frac{36 - 4(4+h)}{3h(\sqrt{4+h+5})(6+2\sqrt{4+h+5})}$$

$$= \lim_{h \rightarrow 0} \frac{36 - 36 - 4h}{3h(\sqrt{4+h+5})(6+2\sqrt{4+h+5})} = \lim_{h \rightarrow 0} \frac{-4}{3(\sqrt{4+h+5})(6+2\sqrt{4+h+5})} = \frac{-4}{3(\sqrt{9})(6+2\sqrt{9})} = \frac{-4}{9(6+6)} \quad (4)$$

$$= \frac{-4}{9(12)} = \frac{-4}{108} = \frac{-1}{27}$$

$$y = \frac{-1}{27}x + b$$

$$\frac{2}{3} = \frac{-1}{27}(4) + b$$

$$\frac{18}{27} = \frac{-4}{27} + b$$

$$\therefore y = \frac{-1}{27}x + \frac{22}{27}$$

APPLICATION

1. A dead branch breaks off a tree located at the top of an 80 m cliff. After time t , in seconds, it has a fallen distance, d , in metres, where $d(t) = 80 - 5t^2$, $0 \leq t \leq 4$.

a) Determine the average rate of change of the distance the branch falls in the first 3 seconds.

$$d(0) = 80 - 5(0)^2 = 80 \text{ m} \quad d(3) = 80 - 5(3)^2 = 80 - 45 = 35 \text{ m}$$

$$\text{AROC} = \frac{35 - 80}{3 - 0} = \frac{-45}{3} = \boxed{-15 \text{ m/s}}$$

b) At what time is the instantaneous rate of change in distance -35 m/sec ?

$$d'(t) = \lim_{h \rightarrow 0} \frac{d(t+h) - d(t)}{h} = \lim_{h \rightarrow 0} \frac{80 - 5(t+h)^2 - 80 + 5t^2}{h} = \lim_{h \rightarrow 0} \frac{-5(t^2 + 2th + h^2) + 5t^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-5t^2 - 10th - 5h^2 + 5t^2}{h} = \lim_{h \rightarrow 0} \frac{-10th - 5h^2}{h} = \lim_{h \rightarrow 0} -10t - 5h = -10t$$

$$-35 = -10t$$

$$t = \frac{35}{10} = \frac{7}{2} = \boxed{3.5 \text{ seconds}}$$

2. If $\lim_{x \rightarrow a} [f(x) + g(x)] = 2$ and $\lim_{x \rightarrow a} [f(x) - g(x)] = 1$, find $\lim_{x \rightarrow a} [f(x)g(x)]$.

$$\begin{aligned} \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) &= 2 \\ \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) &= 1 \end{aligned}$$

$$\lim_{x \rightarrow a} f(x) = 2 - \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} f(x) = 1 + \lim_{x \rightarrow a} g(x)$$

$$2 - \lim_{x \rightarrow a} g(x) = 1 + \lim_{x \rightarrow a} g(x) \rightarrow 1 = 2 \lim_{x \rightarrow a} g(x)$$

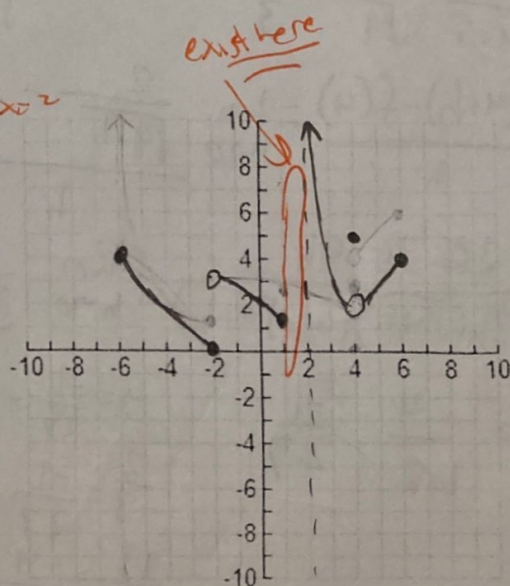
$$\lim_{x \rightarrow a} g(x) = \frac{1}{2}$$

$$\lim_{x \rightarrow a} f(x) = 2 - \frac{1}{2} = \frac{3}{2}$$

$$\lim_{x \rightarrow a} f(x)g(x) = \frac{3}{2} \cdot \frac{1}{2} = \boxed{\frac{3}{4}}$$

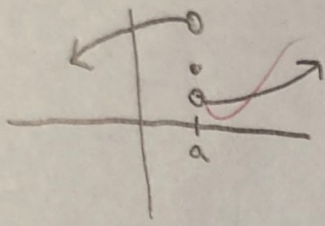
3. Sketch the graph of a function $y = f(x)$ that satisfies the following conditions:

- $\lim_{x \rightarrow 4^+} f(x) = 2$, $\lim_{x \rightarrow 4^-} f(x) = 2$, $f(4) \neq 2$
- Increasing on $4 < x < 6$
- Decreasing on $-6 < x < -2$, $-2 < x < 4$
- Domain: $\{x \in \mathbb{R}, -6 \leq x \leq 6\}$ Need to exist at $x=2$ And $1-2$
- Range: $\{y \in \mathbb{R}, y \geq 0\}$
- Jump Discontinuity at $x = -2$



COMMUNICATION

1. Provide a diagram where $\lim_{x \rightarrow a} f(x)$ does not exist, $f(a)$ is defined and $f(x)$ is not continuous at a . What type of discontinuity is represented in this function?



This is a jump discontinuity

(2)

2. Using proper terminology and formulae, explain in detail the difference between algebraically determining the slope of a secant vs the slope of a tangent.

Slope of Secant $\rightarrow m = \frac{y_2 - y_1}{x_2 - x_1}$
This is like finding the slope of a regular linear

Finding 2 points on the graph and determining the distance between them. Secant ?

(3)

Slope of tangent is like slope of secant, but putting the second point on top of the first point so that it touches the graph once.

To move this point onto $(x, f(x))$, h would have to "approach 0", then plug

$(x+h, f(x+h))$ in these points into the same formula, except with $\lim_{h \rightarrow 0}$.
$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{x+h - x} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$
 this is how you get the formula for slope of tangent.

how?

THINKING

1. Determine the value of the constants a and b if $\lim_{x \rightarrow -1} \frac{2x^2 - ax - 14}{(x-b)(x+1)} = 4$.

The $(x+1)$ must cancel out, therefore it is one of the factors of $2x^2 - ax - 14$.
(2 numbers that multiply to -28 , one is 1 , meaning 2nd is -28 , $a = -(1 - 28) = 27$

$$\lim_{x \rightarrow -1} \frac{2x^2 + 27x - 14}{(x-b)(x+1)} = \lim_{x \rightarrow -1} \frac{(x-28)(x+1)}{(x-b)(x+1)} = \lim_{x \rightarrow -1} \frac{(x-28)}{(x-b)} = \frac{(-1-28)}{(-1-b)} = \frac{-29}{-1-b} = 4$$

Now we can find b

$$\frac{-29}{-1-b} = 4$$

$$a = 27$$

$$b = \frac{25}{4}$$

$$\begin{aligned} -29 &= 4(-1-b) \\ -29 &= -4 - 4b \\ -25 &= -4b \\ b &= \frac{25}{4} \end{aligned}$$

But b/c it's $-a$, a must equal the sum of those 2 numbers multiplied by negative 1.